

Jake van Halder

jakevhbusiness@gmail.com | [portfolio](#) | github.com/jakevanhalder

Skills

Languages: TypeScript, JavaScript, Golang, C/C++, C#, Python

Tools: React, Vite, Git, Linux, Figma

Experience

Software Engineer Intern, Schweitzer Engineering Laboratories – Pullman, WA April 2025 – Present

- Develop enterprise-grade web applications using React (TypeScript) and Golang, improving performance and maintainability of internal and external tools.
- Built a **Jenkins artifact pipeline** using Shell scripting and Node.js to aggregate SEL device data into static HTML dashboards, reducing data retrieval time.
- Implement end-to-end testing pipelines with Playwright and Vitest, increasing test coverage.
- Collaborate in an Agile team environment, contributing to code reviews and CI/CD workflows.

Research Aid 2 (Frontend Developer), Washington State University – Pullman, WA October 2024 – Present

- Lead frontend development of a GIS-based watershed analytics platform supporting hydrologic research across Washington and Oregon.
- Build interactive geospatial features using React (TypeScript) and Leaflet, enabling real-time watershed visualization and data exploration.
- Integrate frontend components with a Django backend to dynamically render watershed properties and time-series datasets.
- Optimize map rendering and spatial data handling for improved performance with large shapefiles.

Projects

Zuno | React Native (Expo), TypeScript, FastAPI, PostgreSQL, Gemini API

- Lead frontend architecture and application structure for a cross-platform React Native (Expo) mobile app, establishing scalable component patterns and state management strategies.
- Design low- and high-fidelity wireframes in Figma, translating product requirements into intuitive UI/UX flows and production-ready interfaces.
- Collaborate with backend (FastAPI + PostgreSQL) and AI (Gemini API) leads to integrate dynamic AI-generated study decks, file uploads, and personalized quiz generation.
- Coordinate frontend development in an Agile workflow using Git and GitHub, contributing to sprint planning, feature reviews, and cross-functional technical decisions.

CGDJ-BulletHell | C#, MonoGame, Git

- Lead a 4-person team in developing a 2D bullet-hell shooter inspired by Touhou Project using MonoGame.
- Architected the project's modular file structure, separating game actors, combat systems, patterns, and stage logic for scalability and maintainability.
- Managed project workflow using GitHub Issues and pull requests, conducting code reviews and coordinating feature integration.
- Established development standards and task templates to streamline collaboration and maintain consistent code quality.

Education

Washington State University – BS in Computer Science

Club – Software Development Club

Expected Graduation: May 2026

September 2024 - Present